

SOLIBERINE®

360° PHOTO-PROTECTOR

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PHYTOBIOACTIVE

Excessive exposure of skin to **UV, visible light and IR** leads to the formation of free radicals, inflammation, apoptosis and ExtraCellular Matrix (ECM) degradation. These are the main causes of premature skin ageing as well as various skin disorders, including short-term (sunburns, dryness, sensitive skin, erythema...) and long-term consequences (dark spots, uneven complexion, photoageing...).

It is now clear established that protection against UV radiations alone is not sufficient enough to prevent skin disorders. In fact, approximately 50% of free radicals that are formed in the skin by solar radiation originate **from the visible light including blue light and infrared rays**. Besides, harmful effects of blue light from artificial sources (smartphones, screens... which are more concentrated in blue light than natural light) become an important issue for public health and skin ageing.

> ACTIONS

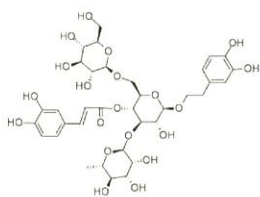
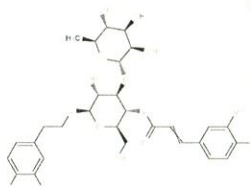
360° PHOTO-PROTECTOR, that prevents premature photoageing:

- Reduces free radicals & Protects skin barrier integrity
- Limits acute and chronic inflammation
- Inhibits hyperpigmentation
- Prevents extracellular matrix degradation

COMPOSITION

Verbascoside: 15%*

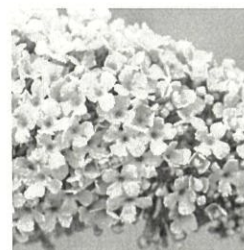
Echinacoside: 3%*



* % by Dry Matter

ORIGIN

Selected for its exceptional richness in **phenylpropanoids**, the purified fraction (caffeic acid free) of flowers from *Buddleja officinalis* is highly concentrated in **verbascoside** and in **echinacoside**, which are responsible for the **biological effectiveness**.



> COSMETIC BENEFITS

- Maintains the normal skin physiology by protecting it against harmful effects of light radiations.
- Fights effectively against ROS production, inflammation, irritation and dryness.
- Prevents occurrence of dark spots and premature wrinkles.
- Preserves skin youth, quality and radiance.

> COSMETIC USES

- ▶ **Daily photo-protection**
- ▶ **Anti sensitive and reactive skin**
- ▶ **Anti-photoageing**
- ▶ **Prevention of redness and skin irritation**
- ▶ **Anti-dark spots**

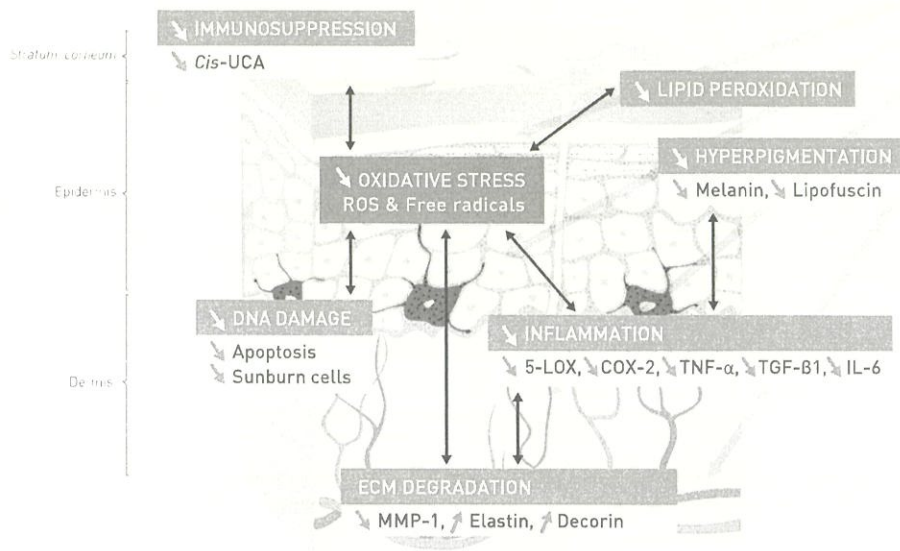


APPROVED DATE... 19 JAN 2020

> MECHANISM OF ACTION

SOLIBERINE®
360° PHOTO-PROTECTOR

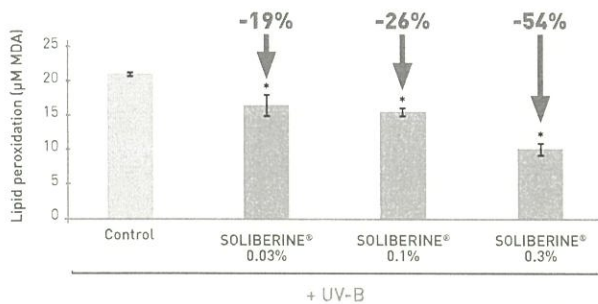
> UV Rays
> Blue Light
> Infra Red



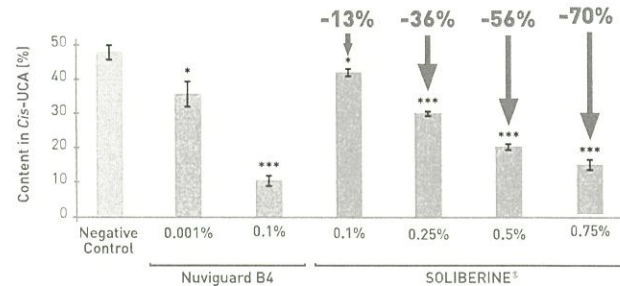
> IN VITRO EVALUATION OF SOLIBERINE ON UV EXPOSURE

• PROTECTION OF SKIN BARRIER

- Inhibition of lipid peroxidation



- Inhibition of *trans*-urocanic acid isomerization

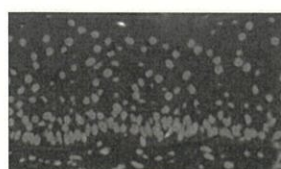


• PROTECTION FROM CELL DEATH

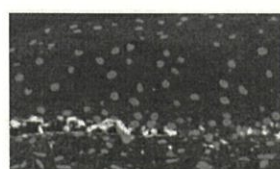
- Inhibition of apoptosis and sunburn cells, on full-thickness skin model

Vs UV-exposure	SOLIBERINE® 0.03%	SOLIBERINE® 0.1%
CELL APOPTOSIS	-45%***	-57%***
SUNBURN CELLS	-26%*	-40%**

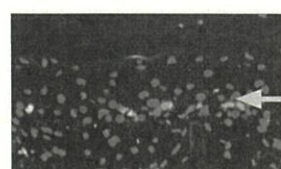
On 3D reconstructed full-thickness model irradiated with UV-B. Quantification of apoptotic cells and sunburn cells after 24h.



Control



Control



SOLIBERINE® 0.1%

Apoptotic keratinocyte
within the basement
membrane

+ UV-B

*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001

• PROTECTION FROM EXTRACELLULAR MATRIX DAMAGES

ELASTIN (SOLIBERINE® 0.1%)

+37%*** Without UV-B

+7%* With UV-B

MMP-1 (SOLIBERINE® 0.03%)

-45%*** Without UV-B

-30%* With UV-B

On full-thickness skin model

• PROTECTION FROM INFLAMMATION-INDUCED HYPERPIGMENTATION

Vs UV-exposure

SOLIBERINE®
0.03%

TNF-α

-36%*

MELANIN

-62%***

On full-thickness skin model

Melanin synthesis



No treated control



SOLIBERINE® 0.03%



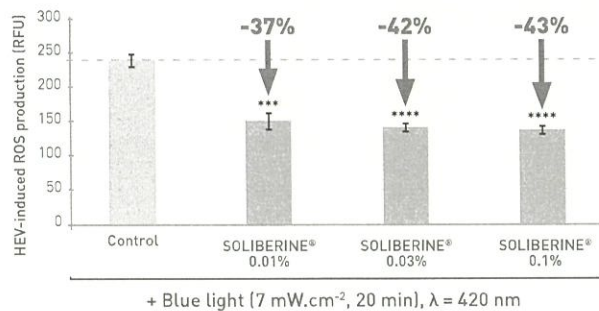
SOLIBERINE® 0.1%

+ UV-B

> IN VITRO EVALUATION OF SOLIBERINE ON BLUE LIGHT EXPOSURE

• PROTECTION FROM OXIDATIVE STRESS

- Inhibition of ROS production, on human keratinocytes



• PROTECTION FROM INFLAMMAGING

- On human fibroblasts, protection from inflammation (TGF-β1) and enhancement of extra cellular matrix (Decorin)

Vs Blue light exposure	SOLIBERINE® 0.03%	SOLIBERINE® 0.06%
TGF-β1 (8H)	-62%***	-88%***
DECORIN (24H)	+18%*	+29%*

• DECREASE OF AGEING SPOTS

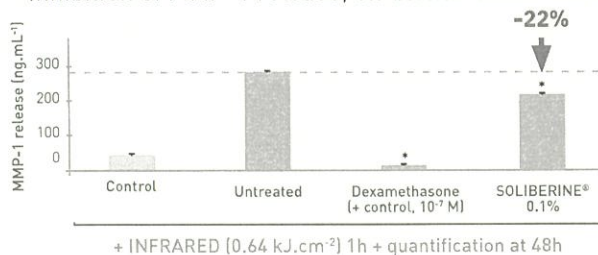
- Inhibition of lipofuscin expression, on dermal fibroblasts



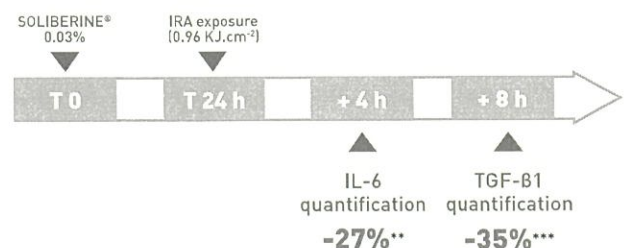
> IN VITRO EVALUATION OF SOLIBERINE ON INFRA RED EXPOSURE

• PROTECTION FROM ECM DEGRADATION

- Inhibition of MMP-1 release, on dermal fibroblasts



• PROTECTION FROM INFLAMMATION ON HUMAN FIBROBLASTS



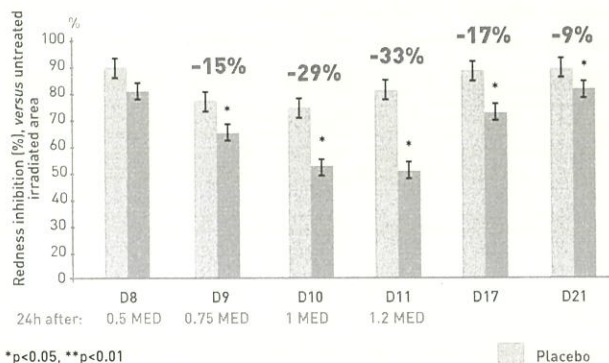
> IN VITRO TESTS

UVA and UVB TEST

Double blind study with 31 female volunteers, phototypes II&III, applied twice daily for 21 days a cream containing 2% SOLIBERINE® on the back (treated zone) versus placebo and untreated zone. Induction of erythema by 4 UV exposures from D7 to D10 with 0.5, 0.75, 1.0 and 1.2 MED respectively. L*a*b* measurements, Individual Typological Angle (ITA°) calculations.

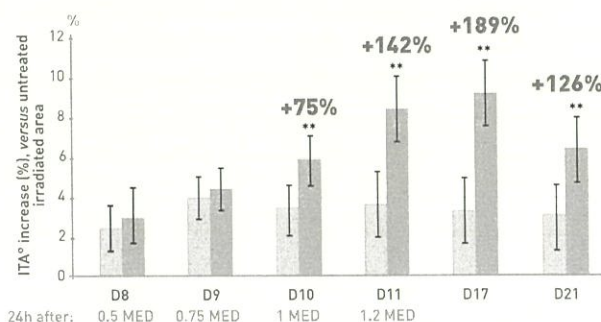
• REDUCTION OF UV-INDUCED ERYTHEMA

Measurement of the skin red colour (a* parameter)



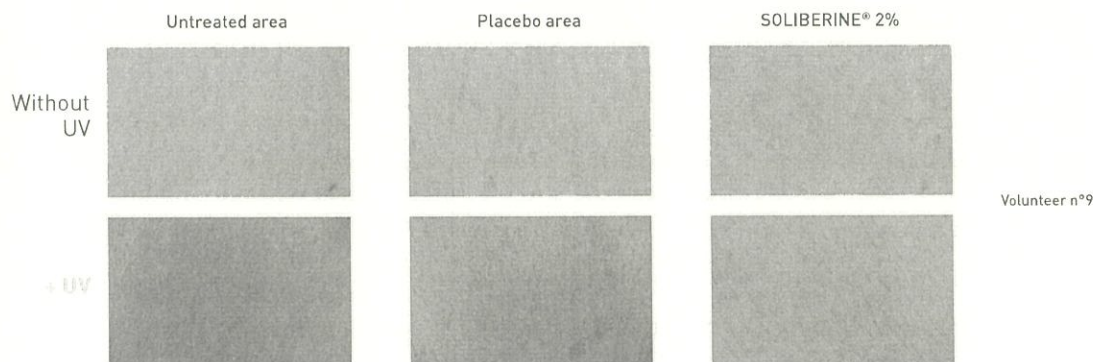
• REDUCTION OF UV-INDUCED PIGMENTATION

Measurement of the skin pigmentation (ITA° parameter)



• VISIBLE EFFECT AGAINST UV-INDUCED ERYTHEMA

The Volunteer skin colours at D11: 24h after the last irradiation at 1.2 MED



SOLIBERINE® - 360° PHOTO-PROTECTOR

A triple powerful skin defense system which maintains skin's youth, quality and radiance.

FORMULATION

Use concentration: 0.5-2%

Use pH: 5.0 - 6.0

Caution for use:

Add at the end of preparation while cooling or at 25-35°C for emulsions.

TECHNICAL DATA

Characteristics

Organoleptic Appearance: liquid
 Colour: Light amber to brown

Solubility Water: Soluble (10%)
 Ethanol: Partially soluble (10%)

Storage

Keep in a dark place in the original packaging and at room temperature (15-25°C).

Tests of tolerance

- Skin irritation: very good skin compatibility
- Eye irritation: slightly irritant
- Sensitisation: very good skin compatibility
- Mutagenicity: not mutagenic
- Phototoxicity: not phototoxic

INCI Name: Buddleja Officinalis Flower Extract / Preservative: Several versions available / Authorized: **

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BIOTECHNOLOGIES
GREENTECH

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